Proposal 4: MLOps / Practical ML (Case Study)

- **Session Title:** From FASTA to Transformer: How I Trained a DNA Language Model on Cheap Laptop
- **Domain:** MLOps / Practical ML (or Data Science / ML)
- **Level:** Beginner / Intermediate
- **Talk Format:** 15-minute / 30-minute Talk
- **Description (Abstract):**

When you think of language models what comes to mind are massive GPU clusters, and companies like OpenAI, Antropic and Nvidia. But what if I tried to train a domain-specific model on my own PC?

In this talk, I will show how I built and trained a DNA language model from scratch on an M2 MacBook with only 8 GB of memory. Rather than competing with state-of-the-art models, I asked a different question: how far can I push a language model when memory, not compute, is the main constraint? And what can I learn along the way.

I will discuss design decisions for the project structure, DNA tokenization and dataset preparation, working around memory limits, evaluation methods and an unexpected shift from a single model to a small ecosystem of specialized models that generate, critique, and guide each other. We will see where my model succeeds, where it fails and especially how I finally got my model to learn how to stop talking.

The audience might leave or leave learning from my journey. How by treating hardware limitations as an opportunity, I ended up learning. You will see my pipelines, infrastructure, what small-scale genomics models can achieve and maybe even have some fun.